Relacks: Reliable Backscatter Communication in Indoor Environments

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The increasing number of embedded, plugged-in radios around homes and offices in everyday objects such as appliances, TVs, and smart speakers provides an excellent opportunity to bring ultra-low-power backscatter connectivity to billions of devices. However, backscatter links suffer from high loss due to two consecutive propagations and have to operate on narrow link-budget margins, which makes them more susceptible to multipath losses in indoor environments. To address this, we propose a closed-loop backscatter system that exploits diversity sources such as communication frequency and transceivers antennas based on the channel metrics to deliver reliable coverage over an entire area. We prototype a backscatter system with Bluetooth Low Energy (BLE) transceivers and BLE compatible tags and deploy it in several multipath rich indoor environments. Our evaluations show that we can successfully communicate with a backscatter tag in a $50m^2$ indoor area. The proposed algorithm for selecting communication parameters achieves an average 2.7 × success rate compared to the random selection while satisfying FCC output power requirements for frequency hopping transceivers.

CCS Concepts: • **Hardware → Wireless devices**; *Sensor applications and deployments*; • **Computer systems organization** → *Sensor Networks*; Embedded systems.

Additional Key Words and Phrases: Internet of Things, Backscatter communication, Reliability, Bluetooth, Indoor, Diversity

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1 INTRODUCTION

The need for low power communication methods is rapidly growing in recent years to satisfy the rising demand for wireless data communication and enable the vision of the Internet of Things [44]. Backscatter communication is known as an ultra-low-power form of communication, where a device preserves energy by reflecting RF signals in the environment instead of generating them. The backscatter device modulates the reflection by modifying the impedance connected to its antenna. Although commercial RFID tags use the same principle, the high cost of the RFID reader deployment and infrastructure has limited the backscatter usage in many applications. Recent works on bistatic backscatter communication have shown great promise on turning everyday radios into

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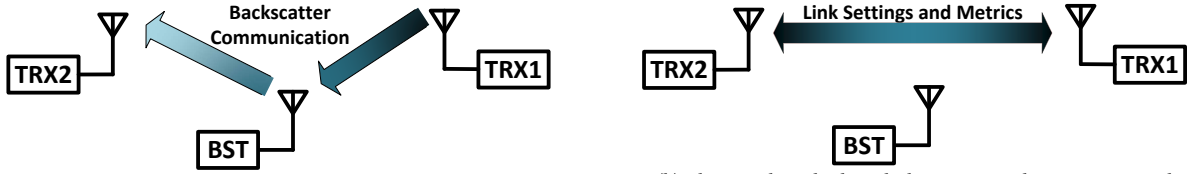
Fig. 1. Our vision for future connected indoor environments, where plugged-in devices use the same embedded radios to connect to the home network and to enable backscatter communication with the smaller objects [8].

backscatter enabled devices by achieving compatibility with different wireless standards [9, 17, 25, 29, 32, 38, 47, 52, 55]. Backscatter communication systems can achieve high throughput [23, 41, 56] and has been successfully demonstrated in applications such as wireless video streaming [27, 43], indoor localization [33, 34, 37], human activity recognition [10, 12, 42] and brain interfaces [41].

The recent wave of wirelessly connected devices in homes provides an excellent opportunity to bring low-cost connectivity to billions of objects around us. For example, the embedded transceivers in the dishwasher, oven stove, and range hood could collaborate and form backscatter links to communicate with smaller or non-plugged objects around the kitchen -such as a cooking pot, a kitchen knife, or cabinets- equipped with low-cost backscatter tags. Fig. 1 depicts this vision where the backscatter communication helps to realize new applications such as reading the temperature inside the cooking pot, detecting human interactions with different objects around the kitchen, and many more. We need to satisfy two main requirements to realize this vision. First, the backscatter communication should use the same transceiver(TRX) as the embedded radio to minimize the additional cost. Second, the backscatter system should reliably cover every point in the room by overcoming frequency and spatial dependent fading in indoor environments.

Previous works in backscatter communication have shown backscatter systems with physical layer compatibility to current wireless standards and measured the maximum communication range and throughputs of their implementation. However, methods and techniques to evaluate and improve bistatic backscatter systems reliability, especially in dynamic multipath rich indoor environments with moving objects, is less investigated. Backscatter systems suffer from high propagation loss due to two consecutive propagations and have to operate on narrower link budget margin compare to active radios. Furthermore, increasing RF excitation power does not necessarily improve the communication reliability of bistatic backscatter systems, since it increases the backscatter signal strength and excitation interference to the receiver at the same time. Thus, selecting proper communication parameters is vital for achieving reliable backscatter communication.

One of the challenges in addressing multipath fading in a bistatic backscatter system arises from the distributed nature of these systems. The transceiver that provides the excitation signal can not detect whether a backscatter packet is formed and received or not. Only the receiver can determine the link quality based on the strength of the reflected signal. To solve this issue, we present *Relacks*, a closed-loop system where the link quality metrics



(a) The TRX boards follow the defined settings to communicate with the tag.

(b) The TRX boards share link settings and metrics to avoid using low quality links for future backscatter communications.

Fig. 2. The transceivers share link settings and metrics after the backscatter communication to improve link reliability.

are shared between the two base units and used to avoid low-quality links, as shown in Fig. 2. We achieve high reliability by exploiting frequency, antenna, and wake-up source diversity in real-time with the optimum communication parameters selected based on the history of the previous communications.

To investigate the effects of closed-loop parameter selection on the backscatter link performance, we design and prototype a backscatter system that uses BLE backscatter nodes to communicate with commercial off the shelf BLE transceivers. Backscatter tags are capable of generating BLE v5.0 coded PHY packets to achieve a better sensitivity and communication range. The tags use the backscatter link to transmit data to base units and are equipped with ultra-low-power wake-up receivers to receive command and data from the base units. BLE TRX base units are designed with off the shelf components and can generate wake up patterns to activate and communicate with the tags. We test the *Relacks* backscatter communication system in multiple indoor environments. The results show that the system achieves a uniform coverage for any point in several indoor environments with areas up to 60 m^2 while the excitation power is 18 dBm. We evaluate the system not only when the tag lies between the two TRXs, but also when the tag is placed on the corners of an area, which are the worst-case arrangement for covering a room.

The main technical contributions of this work are:

1) We propose a closed-loop backscatter system, where link parameters are shared between the two TRXs to improve link reliability. We show the effect of frequency, antenna, and wake up source diversity on the backscatter link reliability and implement a real-time algorithm to enhance communication reliability by using these diversities effectively.

2) We prototype a complete backscatter system with backscatter tags and BLE transceiver units and use them in our evaluations. We design tags with backscatter packets compatible with BLE 5.0 coded and 1M PHYs for uplink and ultra-low-power OOK based packets for downlink. We design transceiver units with commercial BLE transceivers, each one is capable of communicating with the tag, generating RF excitation signal, and receiving backscatter packets.

3) We evaluate the system performance intensively at multiple locations in several indoor environments. Also, we characterize the effect of several variables such as tag antenna orientation and different human interactions on the system performance.

2 BACKSCATTER SYSTEM DESIGN

The first step in designing a backscatter system is to select the communication standard. We target smart home applications, and we would like to use the same transceiver used in connecting to the smart home network for communicating with backscatter tags. Several standards, such as Zigbee, Z-Wave, Thread, BLE, and WiFi, are competing for market share with no one dominating the market at the moment [15, 36]. Bluetooth is a major contender in the battle for smart home protocol dominance with more than 4 Billion devices shipped in 2019 [5], high compatibility between devices, and availability in smartphones. Bluetooth v5.0 has introduced improved mesh capabilities, long-range, and high data rate modes to address some of the shortcomings of the previous

versions. While we have chosen to use BLE for this implementation, the techniques that we describe in section 3 can be applied to any backscatter system to improve the reliability.

To increase the communication range of a wireless system, one can increase transmit power. Increasing the RF source power increases the interference to other users and limits the overall throughput. By limiting the scope of the backscatter communication to a room, we do not need to overcome the signal loss through the walls, and we can lower the RF source excitation power to minimize the interference to the other wireless systems in the area. Since floor plan and room sizes of each house can be very different [7, 46], we target to cover a 50 m^2 area with the backscatter system in this work. The coverage would be enough for most bedrooms, bathrooms, laundry rooms, kitchens, dining rooms, and living rooms in homes less than 3000 ft^2 (278 m^2) [16].

2.1 Bluetooth Low Energy v5.0 Primer

BLE v5.0 standard defines two modulation schemes, a mandatory 1MSym/s and an optional 2MSym/s. Although the 2Msym/s achieves a higher data rate, we stick to the mandatory one since the data rate is enough for most IoT applications, and it consumes less power due to operating with a lower clock rate. The 1Msym/s modulation scheme uses a binary frequency modulation, where a binary '1' represented by a positive frequency deviation and a binary '0' represented by a negative frequency deviation. The minimum frequency deviation shall never be less than 185 KHz at 1Msym/s. The maximum output power level for BLE v5.0 devices is 20 dBm [1].

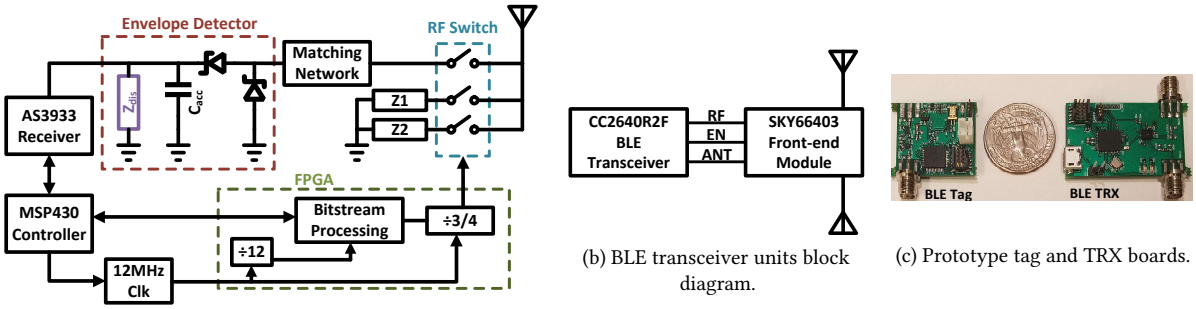
The 1MSym/s modulation supports two PHYs, LE 1M uncoded PHY with 1Mbps data rate, and LE coded with 125kbps or 500kbps data rates. While both PHYs perform CRC generation and whitening on the incoming bitstream, the coded PHY takes two additional steps of FEC encoding and pattern mapping. FEC encoding uses a non-systematic, non-recursive, rate 1/2 code with constraint length equal to 4 to map each incoming bit to two bits. Pattern mapper is only used in the 125kbps coded PHY and converts each output bit of the FEC encoder to four bits, '0011' for zeros and '1100' for ones [1]. By performing these additional coding steps, the coded PHY receivers achieve lower sensitivity (i.e., longer communication range) and better interference handling at the expense of lower data rates. In this work, we focus on the 125 Kbps coded PHY and 1Mbps uncoded PHY.

2.2 BLE Tag

We design the tag to be a complete transceiver with a BLE v5.0 compatible backscatter uplink and an ultra-low-power amplitude modulation based downlink. The uplink and downlink share one antenna using time-division multiplexing. In receive mode, the antenna is connected to the receiver through a SKY13317 three-to-one RF switch. Once a wake-up request is detected, the receiver enables the controller. The controller is a low power MSP430FR2311 microcontroller (MCU), and it controls the backscatter uplink. In transmit mode, the antenna switches between the other two ports of the RF switch, where one connected to GND through a 47pF capacitor and the other one left open. The block diagram of the tag and the fabricated prototype board are shown in Fig. 3(a) and Fig. 3(c) respectively.

BLE5 Compatible Backscatter. The backscatter uplink includes an ICE40UP low power FPGA and a 12 MHz SiT8021 MEMS oscillator. The FPGA and the oscillator remain in an idle mode until the controller enables them. Once enabled, the FPGA receives the data from the MCU and performs all the packet processing steps required for BLE packet generation in coded or uncoded PHYs. The bit generation logic operates at a 1MHz clock rate and outputs one bit each microsecond. The 1MHz clock is generated inside FPGA by dividing the input clock to 12.

To mitigate the RF source interference effect on the receiver, we should modulate the backscatter data with a subcarrier to shift the tag output spectrum respect to the RF source. Considering the 185KHz minimum frequency deviation in the BLE standard, we need to generate two frequencies that are at least 370KHz apart. Higher subcarrier frequency increases the frequency separation between the RF source and backscatter packet, but it also increases the tag power consumption. In this work, we use a 12 MHz reference clock and the lower sideband



(a) BLE tag block diagram.

(b) BLE transceiver units block diagram.

(c) Prototype tag and TRX boards.

Fig. 3. BLE tag and BLE transceiver prototypes.

of the reflected spectrum. Turning the RF switches on and off at 4MHz and 3MHz represents bits '0' and '1' respectively, which means a 3.5 MHz subcarrier frequency modulation and 500 KHz frequency deviation. The two frequencies are generated from the 12 MHz reference clock and satisfy the BLE standard requirements. Since the BLE standard specifies receiver interference performance for adjacent interferers up to 3 MHz [1], increasing the subcarrier frequency beyond 3MHz does not provide any guaranteed performance boost. Furthermore, the selected frequencies are integer multiples of the bit rate, which eliminates any undesired glitch in switching between consecutive bits.

Wake up Receiver. Despite the recent advances in the low power wake up radio architectures [28, 35, 39, 50], there are few commercial-off-the-shelf components available for this task at the moment. We use AS3933 in this work, which is a low-power, low-frequency receiver IC with wake-up pattern recognition. The receiver consumes only 2.3uA in listening mode, which is essential to preserve energy in power harvesting systems or enable multi-year operation on a single battery in battery powered-systems. Each tag is individually addressable using a 16 bit ID. The AS3933 reports the received signal strength (RSS) after each wake up as a number between 0 and 31, and we later use this feature for enhancing system reliability in section 3.

The receiver operates at 125 KHz with a Manchester coded 4.16 Kbps data rate. In Manchester coding, '01' and '10' bit sequences represent symbols '0' and '1' respectively. The incoming RF signal at 2.4 GHz is amplitude modulated with a 125 KHz waveform, which itself is modulated with a Manchester coded 4.16 Kbps data stream. We use a passive envelope detector with SMS7621 RF diodes after the matching network to extract the 125 KHz waveform from the incoming RF signal. The envelope detector circuit diagram is shown in Fig. 3(a). In the presence of the RF signal, the charge accumulates on capacitor C_{acc} and the output voltage rises. In the absence of the RF signal, the capacitor is discharged through Z_{dis} and the output voltage falls. Each charge and discharge cycle should take 8 us to create a 125 KHz waveform.

There is a trade-off in selecting the Z_{dis} resistance. For strong RF inputs, the resistor should be small enough to discharge the capacitor in 4us. For weak inputs, the resistor should be large enough to allow the output voltage to rise during the 4us charge cycle. To solve this issue, we design a non-linear voltage-dependent resistance by combing diodes and resistors, as shown in Fig. 3(a). At lower capacitor voltages, the diodes are off, and only the largest resistor is discharging the capacitor. For stronger inputs, the equivalent resistance decreases as the capacitor voltage rises, and the diodes turn on. We use a 10M ohm resistor, a schotkey diode ($V_{on} = 0.2V$) in series with a 10K ohm resistor, and a general-purpose diode ($V_{on} = 0.7V$) in series with a 1K ohm resistor to form the discharge network. We successfully test the receiver with input powers between 10 dBm and -43 dBm.

Power Consumption To measure the tag power consumption in backscatter mode, we set the tag to generate and send backscatter packets continuously. The total measured power consumption is 1.6mW when the tag transmits 520 coded PHY packets per second. (Each packet is 1.44 millisecond long.) The 12 MHz oscillator is

kept on constantly during the test and consumes 627 uW from a 2.5V supply. The FPGA core consumes 192 uW from a 1.2V supply to generate BLE packets. The FPGA I/Os consume 620 uW from a 2.5V supply to drive the RF switches, and the MCU consumes an average 50 uW from a 2.5V supply to wake up between packets and transfer next packet data to the FPGA. The power consumption in digital circuits is proportional to CfV_{dd}^2 where C is the load capacitance, f is the frequency of the operation, and V_{dd} is the supply voltage. The power consumption of the tag in backscatter mode is dominated by the 12 MHz oscillator and RF switches drive since high-frequency signals are routed on the printed circuit board from one chip to another and experience a large pad and parasitic capacitance. In order to achieve the true low power capabilities of the backscatter communication, all the blocks must be integrated inside one IC to minimize the operating voltage and load capacitance.

IC Design We simulate our BLE backscatter tag design in the TSMC 65nm LP CMOS process using industry-standard EDA tools. Our simulated IC receives payload data and BLE PHY setting and generates a BLE packet by following the packet processing steps as described in 2.1. We use the Design Compiler by Synopsis to convert the Verilog code implemented on the FPGA to transistor-level implementation. We use RF transistors to implement backscatter switches. Our backscatter tag baseband consumes 22 uW from a 1.0v supply to generate BLE packets. We need a 12 MHz reference clock in this design. [29] simulates a ring oscillator based PLL that consume 5.6 uW to generate an 11 MHz clock. However, practical oscillators require additional circuitry to ensure proper operation over process, voltage, and temperature variation. [18] prototypes a 10 MHz oscillator in 65nm, which consumes 23.6 uW in steady-state operation using precisely timed energy injection technique. We estimate a practical 12 MHz oscillator with the same technique consumes 28 uW, which brings the total backscatter tag power consumption to 50 uW. This power consumption is low enough to be harvested from a variety of sources inside a building [54].

2.3 BLE Transceivers

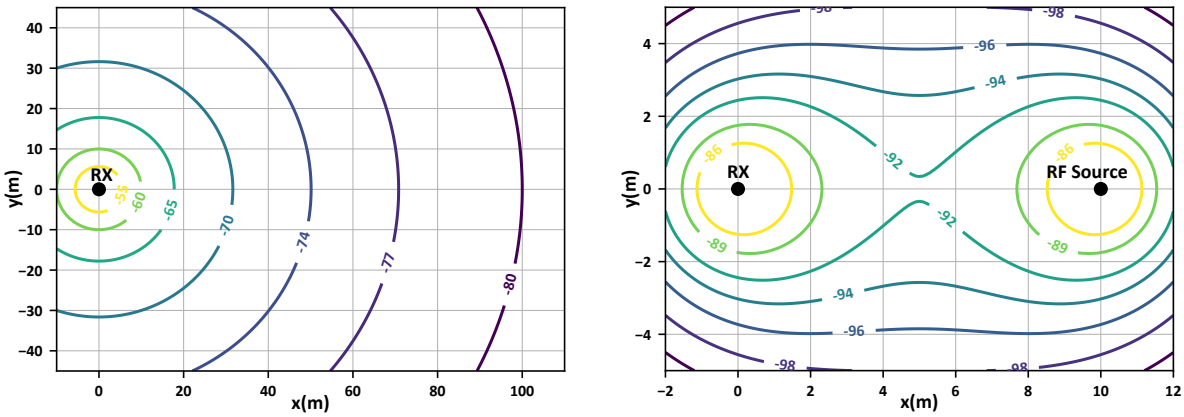
We use commercial-off-the-shelf BLE transceivers in the TRX boards to generate the RF excitation signal or receive packets in backscatter communication. The boards are symmetrical and have both transmit and receive functionalities. The goal is to use the same transceivers to connect to the home network and support backscatter communication, with minimum added components.

Based on the communication protocol and target coverage area, we can calculate the required RF excitation power. Equation 1 expresses the received power at the receiver antenna port in backscatter communication.

$$P_{RX} = P_{TX} + G_{TX} + 20Log(\frac{\lambda}{4\pi d_1}) + CG_{tag} + 2G_{tag} + 20Log(\frac{\lambda}{4\pi d_2}) + G_{RX} - mp_{tx-tag} - mp_{tag-rx} \quad (1)$$

where P_{RX} is the received power at receiver antenna, P_{TX} is the transmitted power at the RF source, $G_{TX}, G_{tag}, and G_{RX}$ are the RF source, tag, and receiver antenna gains respectively, CG_{tag} is the tag conversion gain, λ is the signal wavelength, d_1 and d_2 are the distance between the tag and two TRXs, and mp_{tx-tag} and mp_{tag-rx} are the multipath loss between TRX1-tag and tag-TRX2 respectively.

In this work, we use CC2640R2F BLE transceivers with -103 dBm sensitivity for BLE coded PHY. We assume 0 dBi antenna gain for RF source, tag, and receiver. Tag conversion gain is measured as -4 dB,-3 dB due to dividing the incident power in upper and lower frequency sidebands and an additional 1 dB loss through the RF switch. λ is 0.123 meters at 2.440 GHz. We would like to cover $50m^2$, so $d_1 \times d_2$ is equal to 50. By substituting these numbers in equation 1, while assuming no multipath loss, the minimum required RF source power is 15.3 dBm. Since this output power is beyond the maximum output power of the selected BLE transceiver, we use the SKY66403 BLE front end module to amplify the BLE TRX output power. The front end module supports two antennas, which we discuss how it can be used to improve link reliability in section 3. The front end module requires eight passive capacitors, and the total added cost is less than $1.5 for each TRX board. The TRX board consumes 12 mA from a 3V supply in receive mode and 90 mA in transmit mode when the RF output power is 18 dBm.



(a) Active link with a 0dBm active transmitter.

(b) Backscatter link with 20dBm source power and 4dB loss in tag.

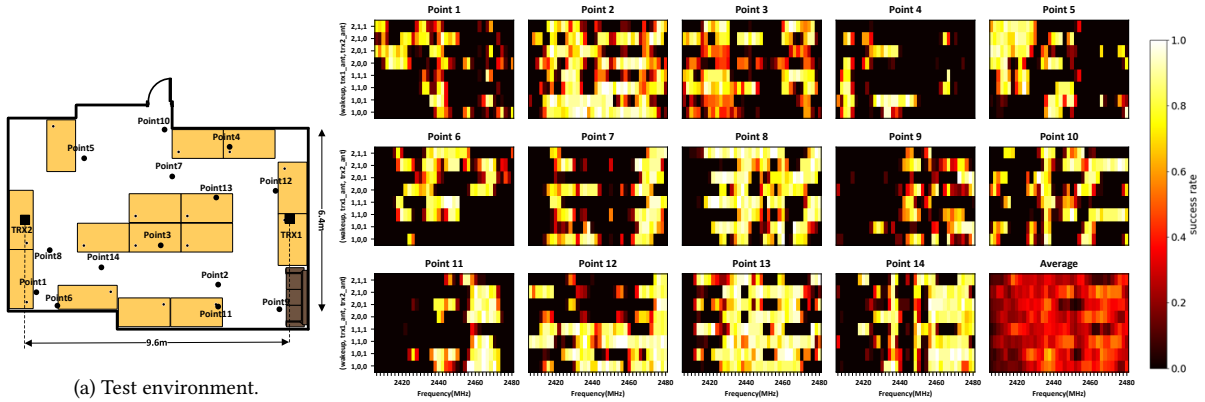
Fig. 4. Comparing received signal strength for an active link and a backscatter link based on the free space path loss model. The backscatter link has to operate on a much narrower margin to receiver sensitivity.

The AS3933 receiver uses amplitude modulation, but the selected transceiver only supports frequency modulation. To generate amplitude modulated waveform with the TRX boards, we turn the front end module power amplifier on and off. We use the CC2640R2F SPI interface to control the power amplifier in order to guarantee accurate timing in generating the 125 KHz waveform. The BLE transceiver also selects which antenna must be activated. The BLE TRX board block diagram, and the prototype BLE TRX board are shown in Fig. 3(b) and Fig. 3(c) respectively.

3 IMPROVING BACKSCATTER RELIABILITY

Active radio links typically enjoy much higher link budget margins compared to the backscatter ones. To better understand this difference, we simulate and compare the received signal strength for an active and a backscatter wireless link using the free space signal propagation model. Fig. 4(a) shows the received signal strength when the transmitter is placed on the contour lines for an active transmitter with 0 dBm output power. If we assume a BLE receiver with better than -95 dBm sensitivity [2, 4] and a transmitter 50 meters away from the receiver, the received signal strength calculated by the free space path loss model is 21 dB above the receiver sensitivity. BLE is known as a short-range standard with a practical range of 10-50 meters, so the additional margin makes the system more reliable against frequency-selective multipath fading and non-line-of-sight scenarios. Fig. 4(b) shows the received signal strength when the backscatter tag is placed on the contour lines based on equation 1 where RF excitation source is placed 10 meters away from the receiver, $P_{TX} = 20\ dBm$, $CG_{tag} = -4\ dB$, all the antennas have 0 dBi gain, and 0 dB multipath loss is assumed. The simulation shows that even with 100 times higher transmit power and five times shorter distance between the RF source and receiver, the backscatter signal power at the receiver is substantially lower than the active radio for most of the points in the environment. We use free space models with no multipath fading for these simulations. However, the propagation from RF source to tag and tag to RX might also experience fading in real scenarios, which further attenuates the received signal power. Thus, it is vitally important to minimize propagation losses to achieve reliable backscatter communication.

Previous studies have looked at the effect of multipath in indoor environments and methods to overcome it for active radios [11, 13, 14, 19, 22, 30, 31, 45, 48]. Multipath fading is frequency- and spatial-selective; for a specific point in the environment, the multipath loss varies in different frequencies. Similarly, for a specific frequency, the loss varies at different points. Thus, we can use frequency and spatial diversity to minimize the multipath effect.



(a) Test environment.

(b) Success rate variation by using different configurations (frequency, antenna, and wake-up source) at different points.

Fig. 5. Diversity effect on backscatter communication success rate.

The backscatter system that we described in section 2 operates over 2.4 GHz ISM band with 80 MHz bandwidth, and each TRX board has two antennas. So the question is how to select the frequency and antennas to avoid high loss channels. Assuming 2MHz frequency spacing between channels similar to the BLE standard, there are 38 frequency options (RF source and receiver frequency both have to be between 2402 MHz and 2480 MHz, and they are 3.5 MHz apart) and four antenna options. Also, we need to select which device is going to wake up the tag before the start of backscatter communication. We call each set of parameters that has to be fixed before the communication a *configuration*. Thus, we have 304 possible configurations in this system.

To gain a better understanding of the channel variation in a backscatter system, we perform the following experiment: We place two TRX boards on two sides of a 6.4 m × 10.3 m room, 9.4 m away from each other, and place the tag on 14 different points inside the room as shown in Fig. 5(a). At each point, we transmit 50 coded PHY backscatter packets with every possible configuration and calculate the backscatter success rate by dividing the ratio of successfully received backscatter packets to 50. The success rate heatmap plots are shown in Fig. 5(b) for each point where each square shows the success rate for one specific configuration. A brighter square means that the channel specified by that configuration has a better quality. The average plot in the bottom right corner shows the average success rate for each configuration at the 14 measured points. It is more uniform than any of the individual heat maps, which highlights the fact that each configuration might have good or poor quality in different points, and there is no single configuration that achieves a high success rate at all the locations.

Frequency Diversity. The frequency diversity is the most important source of diversity that we can exploit to achieve higher reliability. Each row on the heatmaps in Fig. 5(b) demonstrates the effect of frequency diversity while the other parameters are fixed. On each row, some frequencies are entirely dark, which means they experience a high loss, and no packet is received at those frequencies. On the other hand, some frequencies are bright, which means most of the packets are successfully received at those frequencies. The bright and dark squares are packed together in most cases. In other words, although multipath fading is frequency selective, it changes slowly as the frequency changes. The average success rate over the 14 points shown in Fig. 5 is 30.9% with random configuration selection, but it increases to 91.9% by selecting the best frequency in each case.

Antenna Diversity. Antenna diversity is another way to improve the reliability of the communication link. In these experiments, we use omnidirectional dipole antennas with -0.6dBi average gain [3], and the two antennas on each transceiver are fixed 8 cm apart. The channel variation over antenna changes from point to point; for example, points 4, 6, and 7 show higher channel variations compared to points 12, 13, and 14. The importance of antenna diversity is highlighted for points 4 and 6, where one antenna shows poor quality links (TRX1 antenna

1 for point 4, TRX1 antenna 0 for point 6). In contrast, the other antenna has a high success rate over some frequencies. The average success rate over the 14 points further increases from 91.9% to 95.7% by choosing the best frequency and antenna combination.

Wake up Diversity. We have to specify one of the transceivers to wake-up the tag since they will interfere with each other if they both transmit the wake-up pattern. Ideally, we would like the closer transceiver to do the job, but we do not have the tag location information. The effect of wake-up source selection on overall performance is more evident when the tag is closer to one of the transceivers, such as points 4 and 12, where the TRX1 shows a clear advantage for wake up over TRX2. The average success rate over the 14 points increases from 95.7% to 99.2% by choosing the best frequency, antenna, and wake up source.

3.1 Closed-loop Configuration Selection System

In this section, we explain the closed-loop control system that selects a configuration for each backscatter communication. Our system includes one central controller, two TRX boards, and one or several backscatter tags. Each tag has a unique 16 bit ID.

The closed-loop process is shown in Fig. 6. The central controller is a laptop or a single-board computer that can run python scripts. It is connected to one of the TRXs through the UART interface. We name the TRX connected to the controller receiver and the other TRX transmitter. The controller sends a communication request to the receiver. The communication request sets the tag ID, frequency, each TRX antenna, and the wake-up source. Upon the reception of this request, the receiver sends an active BLE packet to the transmitter and shares this information. This packet also helps to maintain time synchronization between the two TRX boards through the rest of the operation. Next, the TRX board responsible for wake-up generates and transmits the wake-up pattern for the specified tag ID, while the other TRX waits. Once the wake-up is transmitted, the backscatter phase begins.

In the backscatter phase, the transmitter sends out the continuous wave excitation signal at its specified frequency, and the receiver is listening for the backscatter packet at its specified frequency. Upon wake up, the tag MCU reads the wake-up RSS from the AS3933 radio and enables the 12 MHz clock and the FPGA. After a 2 ms settling time, the MCU sends out the data, including a two-byte sequence number and the wake-up RSS, to the FPGA and the FPGA creates a BLE packet with the data and transfers it by modulating the impedance connected to the tag antenna through the RF switch. To ensure time synchronization during backscatter operation, the receiver and transmitter keep a 300 us guard interval before the backscatter process starts and after it finishes. Once the backscatter communication is completed, the receiver shares the result with the central controller. The result includes whether the communication was successful or not, the backscatter RSS, the wake-up RSS, the tag sequence number, and the rest of the tag data. If the devices are programmed to transfer more than one backscatter packet per wake-up, this process repeats for a pre-specified number of times. All the delays in receiving wireless signals depend on the internal architecture of the ICs we use in our implementation and are constant over time. We measure these delays accurately and add delay to the operation of other devices wherever necessary to ensure proper timing between all the devices. We successfully receive up to 40 backscatter packets with one wake up without any timing mismatch.

The central controller saves each backscatter result and selects a new configuration for the next backscatter communication phase based on this information. The same process starts again when the controller sends a new communication request to the receiver.

3.2 Real-time Configuration Selection Algorithm

The goal of the configuration selection algorithm is to use the channel metrics to select the configuration with the highest success probability. An ideal configuration selection algorithm should have several characteristics.

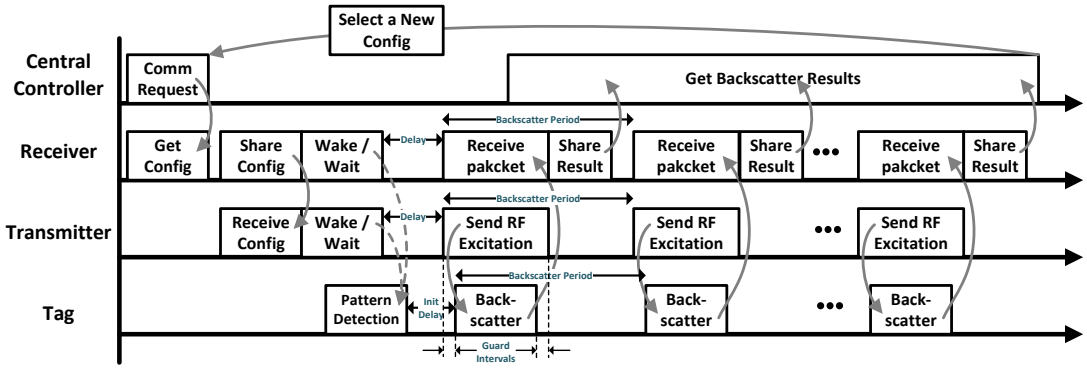


Fig. 6. Closed-loop configuration selection system. Each cycle starts when the controller sends a communication request to the receiver. The TRXs and tag follow the timings to avoid any timing mismatch. The backscatter results are returned to the controller and used to select future configurations.

First, it should find and converge to the optimum configuration with the highest success probability. Second, the algorithm should have a low response time. After the initial start, it should find the best configuration as fast as possible. Finally, it should track environment variations. If a change in the environment happens, for example, a door opens or closes, the algorithm should re-adapt to reflect the change.

In order to find the best configuration, we keep a score vector (S) for each tag over all possible configurations, where each entry is the score for a specific configuration. To assign a score to a configuration, we first need to define a score for one backscatter communication event. We define the backscatter communication event score as equation 2.

$$score = \begin{cases} sqrt(rss_{wake} * rss_{bs}) & \text{if packet is received} \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

where rss_{wake} is the received signal strength reported by the tag wake-up receiver, and rss_{bs} is the backscatter packet received signal strength reported by the receiver minus the receiver sensitivity. The defined score prevents high interference channels by assigning a score of 0 to failed communications. Channels with higher margins from the link sensitivities get a higher score based on both wake-up and backscatter signal strengths.

After assigning a score to each backscatter communication, we update the score for the configuration used in that communication by equation 3:

$$S_{new}[config] = S_{old}[config] * \alpha + score_{bs} * (1 - \alpha) \tag{3}$$

where $S_{new}[config]$ is the new score for the used configuration, $S_{old}[config]$ is the old score, $score_{bs}$ is the backscatter score based on equation 2 and α is a smoothing factor. The smoothing factor controls how many packets the system keeps in memory. For example, it takes 23 dropped packets to cut a score in half if the smoothing factor is equal to 0.97, while it takes only two packets if the smoothing factor is 0.7. Longer smoothing factors keep the previously learned results for longer periods and can handle temporary channel variations; for example, when someone enters a room and leaves it quickly. On the other hand, a smaller smoothing factor is more suitable to respond to permanent channel variations; for example, someone enters a room and sits behind a desk.

To break this trade-off, we keep two separate score vectors with two different smoothing factors and use the same formula as equation 3 to update them separately. We pick one of the two vectors randomly to report the scores. This approach ensures that at least one of the score vectors is on the right track to respond to the change

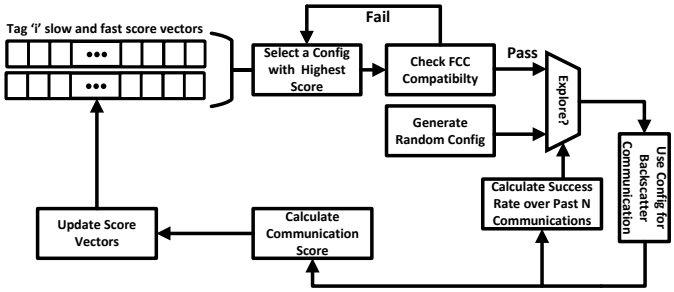


Fig. 7. The proposed configuration selection algorithm. The score vectors keep a ranking of all possible configurations. On each iteration, the configurations with the highest scores are checked for FCC compatibility, and one is selected. The exploration rate is set based on previous communication results and forces a random configuration if exploration is needed.

in the environment, and since both vectors are updated each time with the same backscatter score, they converge to the same scores after a transient period.

Finally, we need to pick a configuration for the next communication based on the tag score vector. Initially, the score vector is filled with zeros, and all configurations have the same score. So, the algorithm picks a configuration randomly until the first successful packet is received, which causes one entry in the score vector to increment from zero. However, there is no guarantee that this is the best possible configuration. Thus, we need to explore other possible high-score configurations. At the same time, we do not want to waste resources by trying low-score configurations over and over again. So, we propose a success rate dependent exploration rate. We calculate the success rate over a fixed-length window of previous backscatter events, and use it as a confidence number for the score vector. A high success rate means that the score vector is suggesting high-quality channels and there is less need for exploration, while a low success rate means that we should keep looking for possibly better configurations. For example, if the success rate is equal to 0, we would like to reject the score vector suggestions altogether and set the exploration rate to 1. On the other hand, if the success rate is 1, we do not need to explore, and the exploration rate should be equal to zero. In this work, we use past 100 backscatter communications to calculate the success rate and define the probability of exploration as $1 - tanh(3 * successrate)/tanh(3)$.

The proposed configuration selection algorithm is implemented in python and runs on the central controller. Fig. 7 shows the algorithm block diagram.

3.3 Frequency Hopping

FCC has specific regulations for the devices operating in the 2.4 GHz ISM band, which limits the maximum output power of devices depending on their operating conditions. A frequency hopping device can transmit up to 0.125W (21 dBm) if it pseudo-randomly switches between 15 non-overlapping hopping channels, and up to 1W (30 dBm) if it uses 75 such channels [6].

Our proposed system picks configurations based on their success rates, and it is likely to suggest the same frequency each time. To solve this issue, we keep a list of the last 15 used frequencies and check the suggested configuration with the highest score against this list. If we have already used the frequency in one of the last 15 tries, we move to the next configuration with the highest score. If configurations with 15 distinct frequencies are suggested, and all rejected because they were in the used frequencies list, we pick the configuration with the frequency equal to the first entry in the list.

Selecting 15 frequencies out of the 40 BLE channels could degrade the system reliability since it forces the system to use channels with higher loss. Based on the FCC regulation, hopping channel carrier frequencies separation shall be equal to the 20 dB bandwidth of the hopping channel or 25 KHz, whichever is greater. Also, the system receivers shall have input bandwidths that match the hopping channel bandwidths [6]. Since our BLE

receiver bandwidth is 1 MHz, we set the hopping channel carrier frequencies separation to 1 MHz. This change increases the number of possible frequencies from 38 to 75 and the number of possible configurations from 304 to 600, which potentially also increases the number of low loss configurations. Our BLE transceivers support much smaller frequency resolution, so this change does not affect the system cost.

3.4 BLE Compatibility

BLE standard uses a pseudo-random hopping pattern of the 37 frequencies in the ISM band. The hopping pattern can be adapted to exclude frequencies that experience high interference to improve co-existence with the non-hopping devices. Once the BLE devices are connected, the channel hopping occurs at the start of each connection event [1].

Unlike BLE that uses frequency hopping to avoid interference from non-hopping devices, the proposed configuration selection algorithm relies on frequency hopping to avoid high loss channels. As discussed earlier, backscatter links suffer from high propagation loss due to two consecutive propagations, and it is vitally important to use channels with a lower loss to make reliable communication. Furthermore, the proposed algorithm is not in a connection state with any of the tags. The backscatter tags stay in low power mode until they are woken up by the TRXs. One could consider using a hopping pattern similar to BLE for periodic backscatter packet exchange in a static environment in a future implementation.

Although the frequency hopping objective and algorithm are different in the BLE standard compared to the proposed backscatter system, our backscatter system uses the same packet format and PHY specifications as the BLE standard. The proposed frequency hopping is implemented using off the shelf BLE transceivers, which shows that it does not require any hardware changes or add to the system cost.

3.5 Synchronization

The transceivers and the backscatter tag have to be synchronized in time and frequency to make a successful backscatter communication. Time synchronization means that the RF excitation signal must be available, and the receiver must be in receive mode before the tag starts to send out its data. Frequency synchronization means that assuming the tag uses a f_{sub} subcarrier frequency, the two TRXs frequencies must differ from each other by the same f_{sub}.

In this work, we use two wireless packets to synchronize the devices before the backscatter phase, as explained in section 3.1. The first one is an active BLE packet from the receiver to the transmitter that shares the configuration information and sets the start of the operation. The second one is the wake-up packet that one of the TRXs sends to the tag to bring it out of low power sleep mode and signal the presence of RF excitation to the tag. We repeat the synchronization process before each backscatter communication phase. This scheme uses the same packets for synchronization and sharing configuration and provides maximum flexibility in modifying the configuration settings for backscatter communication.

The proposed synchronization process takes a fixed duration. The packet between the TRXs and the wake-up packet is 144 us and 4 ms long, respectively. So, the synchronization overhead depends on the length and number of backscatter packets that are transmitted in the backscatter phase. For example, a BLE coded PHY packet with 22 bytes of payload takes 2 ms. Thus the synchronization overhead is 67% if the tag transmits only one backscatter packet, and it drops to 5% if the tag transmits 40 backscatter packets in each wake-up.

In order to reduce this overhead in a future implementation, the synchronization and configuration sharing packets can be separated from each other. The devices can be set to maintain their synchronization by periodically exchanging packets between them once they are synchronized. Such implementation can still benefit from the performance boost resulted from using low-loss channels, but it has to rely on separate packets for that purpose.

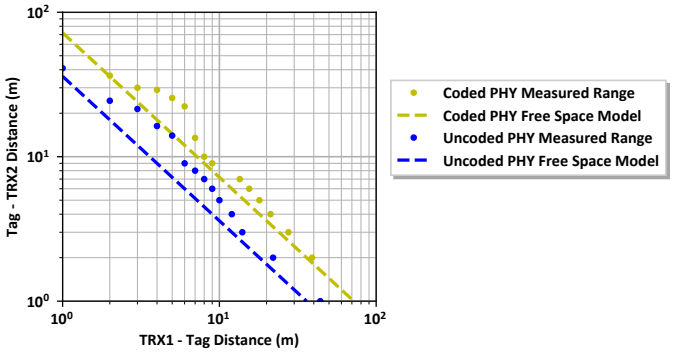


Fig. 8. Communication range of the backscatter system in the line of sight.

In monostatic backscatter systems[24], where one device provides the RF excitation signal and receives the tag reflections, no wireless packet is needed to synchronize the RF excitation source and receiver. These devices could still benefit from the configuration selection algorithm.

4 SYSTEM EVALUATIONS

We evaluate the *Relacks* system in indoor environments with rich multipath reflections. For all the measurements, the tags and transceivers are placed on tripods 1.6m above the ground. We use omnidirectional dipole antennas with -0.6 dBi average gain and maximum 0.5 dBi gain in the X-Y plane [3].

4.1 Communication Range

First, we evaluate the system communication range for the BLE 1 Mbps uncoded PHY and the 125 Kbps coded PHY packet formats. We run the test in a long corridor following the timing described in section 3.1. We set the tag to send one packet upon each wake-up, and the RF excitation power is set to 18 dBm. To measure the maximum range, we fix the TRX1 position and place the tag at different distances from it. Next, we move the TRX2 away from the tag on a straight line until the link is broken. The acceptance criteria for the maximum range is to receive at least 70 percent of the packets successfully over a 5 second period. The results are shown in Fig. 8.

To calculate the free space model base line, we rearrange equation 1 as:

$$Log(d_1) + Log(d_2) = \frac{1}{20}[P_{TX} + G_{TX} - CG_{tag} + 2G_{tag} + G_{RX} + 40Log(\frac{\lambda}{4\pi}) + -mp_{tx-tag} - mp_{tag-rx} - P_{RX}] \quad (4)$$

Everything on the right side of the equation is known in our experiments. The free space models are calculated by placing 18 dBm for P_{TX}, 0 dBi for antenna gains, 4 dB for CG_{tag}, -80 dB for $40Log(\frac{\lambda}{4\pi})$, 0 dBi for mp_{tx-tag} and mp_{tag-rx} and the receiver sensitivity for P_{RX} (-103dBm for coded PHY and -97dBm for uncoded PHY).

The backscatter system achieves an average 42.5m communication range when the tag is 1m away from one of the TRXs using uncoded PHY, and an average 37.5m range when the tag is 2m away from one of the TRXs using coded PHY. Both PHYs follow the trend predicted by the free space model. We observe an average 1.79 × range improvement by using the coded PHY compared to the uncoded PHY, which complies with our expected 2 × improvement due to the 6dB better receiver sensitivity in the coded PHY. Furthermore, we observe that our system achieves a better communication range compared to the free space model baseline at every measured point. We should note that the multipath reflections in indoor environments can combine at the receiver antenna constructively or destructively. A constructive combination of the reflected signals would expand the communication range because the power of the reflected signals adds up together. Since our system dynamically searches for frequencies and antenna pairs that work best, it exploits the constructive multipath to achieve a slightly better communication range than what the free space model predicts.

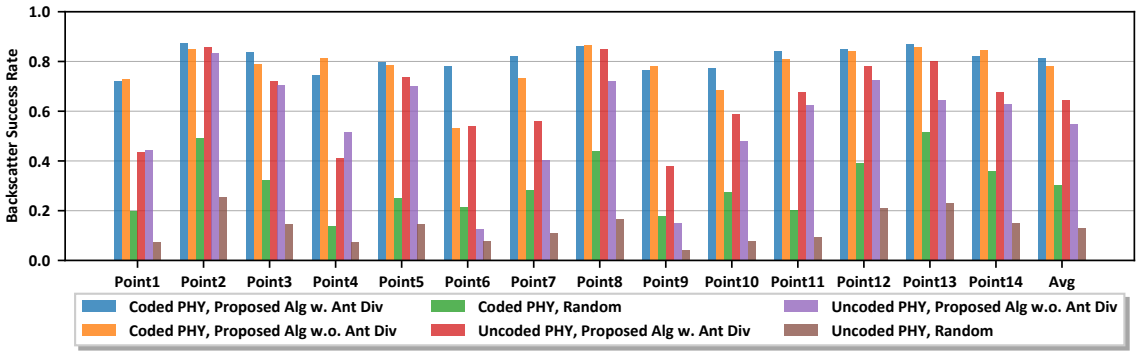


Fig. 9. Backscatter success rate in 14 different points in a 6.4m x 9.4m area for different PHYs and configuration selection algorithms.

Equation 4 shows the clear connection between the RF source output power and the backscatter coverage area ($d_1 d_2$). Reducing the RF source output power by 6dB would effectively reduce the coverage area by half. For example, if we use 0 dBm RF source output power instead of 18 dBm, we expect to achieve 1/8 of the measured maximum range and coverage.

4.2 Backscatter Success Rate

To test the reliability of the system in covering an entire room and to verify the performance of the configuration selection algorithm, we measure the backscatter success rates at the same 14 points shown in Fig. 5(a). At each point, we tried both coded and uncoded PHYs, with three different configuration selection algorithms: random, the proposed algorithm with antenna diversity disabled, or enabled. When the antenna diversity is disabled, we limit both TRXs to one of their antennas. (Left antenna on TRX1 and right antenna on TRX2.) In total, we perform six different measurements at each point. For each measurement, the TRXs are set to communicate with tag 3000 times. We consider the first 500 packets as the start-up period and calculate the backscatter success rate among the last 2500 packets. Each communication follows the timing diagram shown in Fig. 6 with one backscatter packet communicated per cycle. As explained in section 3.5, the successful reception of the backscatter packet relies on the two synchronization packets. In other words, if the active BLE link between the two transceivers, the wake-up link between one TRX and tag, and the backscatter link between the two TRXs and tag have individual packet error rates equal to α, β and γ the overall backscatter success rate would be equal to $(1-\alpha)(1-\beta)(1-\gamma)$. Fig. 9 shows the backscatter success rates for each measurement.

The measurement with the coded PHY packet structure, the proposed configuration selection algorithm, and antenna diversity enabled achieves an average 81.2% backscatter success rate over the 14 tested points inside the room, which shows 2.67 × improvement in success rate compared to random configuration selection. The minimum backscatter success rate is 72.2%, which highlights the uniform and reliable coverage inside the room. The measurement with the coded PHY packet structure, the proposed configuration selection algorithm, and antenna diversity disabled results in an average 78.0% backscatter success rate. It shows similar performance to the case with antenna diversity enabled. The exception is point 6, where the single antenna system achieves only 53.4% compared to 78.3% success rate of the system with antenna diversity. The result shows the importance of using antenna diversity to provide reliable coverage for every location.

Similarly, the measurements with the uncoded PHY packet structure and the proposed configuration selection algorithm achieve an average 61.5% and 53.2% backscatter success rates with and without antenna diversity respectively, compared to 13.1% backscatter success rate for random configuration selection. The uncoded PHY shows a higher variation in success rate, with three locations (points 1, 4, and 9) have success rates below 50%.

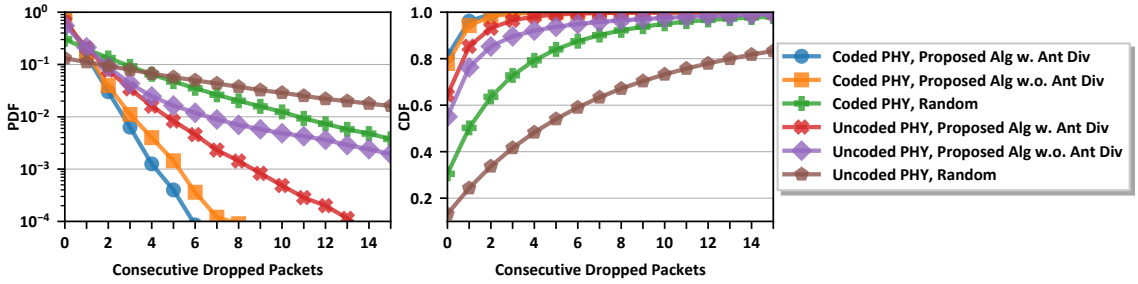


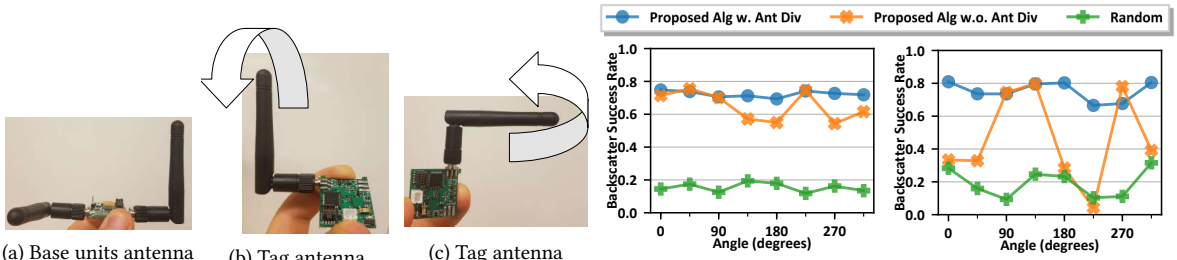
Fig. 10. Probability distribution function and cumulative distribution function for the number of consecutive dropped packets. (Value '0' implies a successfully received packet.)

On the other hand, the uncoded PHY achieves a similar backscatter success rate to the coded PHY at several locations (Points 2, 8, 12, and 13) with eight times data rate. Our current backscatter tag uses a physical switch on the circuit board to switch between coded and uncoded PHYs. However, in a future system, one can send commands from the TRX units to the tags to set the PHY format and trade data rate for reliability.

Consecutive dropped packets. In many applications, we would like to collect the sensor data on fixed intervals, for example, each second or minute. To ensure that our backscatter system could hand over the information on each interval, we not only need to test the system's average success rate but also the distribution of the dropped packets. We analyzed the result of the measurements at 14 different points to gain a better understanding of the dropped packets distribution. For each packet, we counted the number of dropped packets till the next successfully received backscatter packet (For successful packet, the number is equal to zero). The probability distribution function and cumulative distribution function of the consecutive dropped packets for both PHYs and all the configuration selection algorithms are plotted in Fig. 10. The plot shows that the likelihood of receiving a backscatter packet on the first try is 81% for the coded PHY packets with antenna diversity enabled and proposed algorithm. Furthermore, the likelihood of one dropped packet followed by a successfully received packet is 15%, and the likelihood of 5 consecutive dropped packets before a successful packet is less than 0.1%. We can use the CDF plot to calculate the number of required backscatter packets to make sure at least one packet is received with a given confidence number. For example, if the tag sends out three coded PHY packets while the system is using the proposed configuration selection algorithm, higher than 99% likelihood that at least one will be received successfully.

Delay interval effect on success rate. In previous experiments, we have received an average of 70 backscatter packets from the tags each second, which means 14 ms interval between packets. We increase the time interval between packets by introducing 0.1s, 0.3s, and 1s delay intervals on the controller to investigate its effect on the backscatter success rate. The tag and the TRXs are placed at fixed points during the test. The measured backscatter success rates for the coded PHY packets with the proposed configuration selection algorithm are 83.0%, 83.1%, 83.6%, and 84.6% for 14ms, 0.1s, 0.3s, and 1s delay respectively. For the uncoded PHY packets, the backscatter success rates are 65.5%, 69.7%, 68.6%, and 66.3% for similar algorithms and delays between packets. We conclude that the delay interval between packets does not have a direct effect on the backscatter success rate.

Random exploration effect on success rate. Our proposed algorithm relies on random configuration selection to find the configurations with the highest success rates. To investigate the effect of this random exploration on the measured results, we fix the tag and TRXs position and run the same measurement ten times. In each run, we collect 10000 packets. The average backscatter success rate over the ten measurements is 73%, and the standard deviation of the success rates is 2.6%. The low standard deviation implies that our measurements are repeatable, and the random exploration has a limited impact on the overall performance.



(a) Base units antenna orientation for the test.

(b) Tag antenna rotation in the XZ plane.

(c) Tag antenna rotation in the YZ plane.

(d) Backscatter success rate dependence on tag antenna orientation in the XZ plane.

(e) Backscatter success rate dependence on tag antenna orientation in the YZ plane.

Fig. 11. Tag antenna orientation effect on the backscatter success rate.

4.3 Tag Antenna Orientation

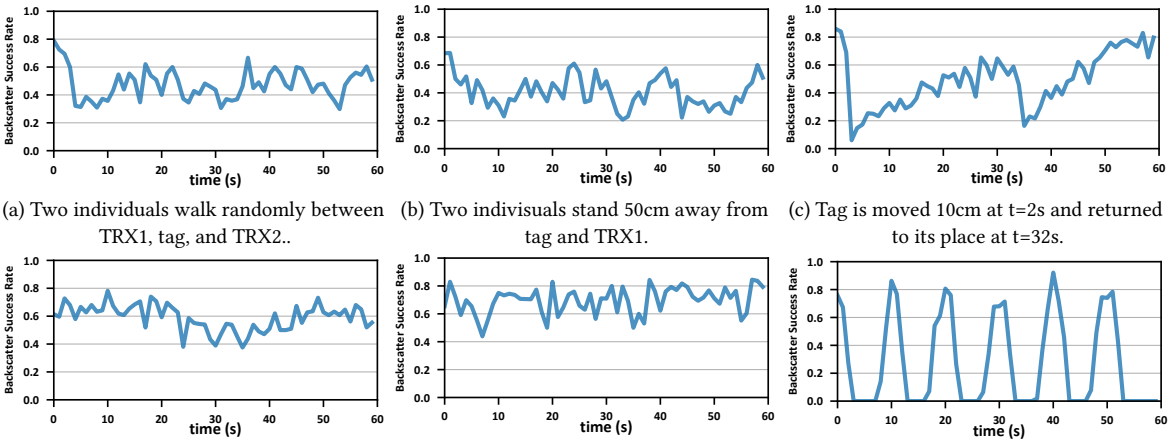
Backscatter tags depend on reflecting the RF excitation signal to transfer their data. Antenna orientation mismatch between the tag and TRXs could affect the RF signal available at the tag antenna, and limit the communication range and reliability. To quantify the effects of the antenna orientation mismatch on our backscatter system, we set both TRXs antennas as shown in Fig. 11(a) and change tag antenna orientation in the XZ and YZ planes in 45 degrees step, as shown in Fig. 11(b) and Fig. 11(c) respectively. At each orientation, we collect 3000 coded PHY packets by following three different settings, as described in section 4.1. The backscatter success rates are shown in Fig. 11(d) and Fig. 11(e) for the XZ and YZ planes respectively. The proposed algorithm with antenna diversity maintains an average backscatter success rate of 72% and 75% in the XZ and YZ planes and show small variation with the tag antenna orientation. However, when we disable the antenna diversity and limit each TRX to one antenna, the backscatter success rate drops substantially at certain orientations. These tests clearly show the importance of antenna diversity in handling random tag antenna orientation.

Unlike frequency diversity and wake-up diversity, antenna diversity requires hardware changes. Increasing the number of antennas increases the number of independent channels and the likelihood of successful communication. However, as the number of antennas increases, the returns from adding one more antenna is diminishing [20]. Our results show that although one antenna on each TRX with frequency diversity can improve the backscatter success rate, a second antenna on each TRX quadruples the number of independent channels and helps to make the system immune to variation in tag antenna orientation. We achieve a higher than 70% success rate at every tested point and antenna orientation, as shown in Fig. 9 and Fig. 11, so we expect marginal improvements by increasing the antennas on each TRX beyond two.

4.4 Environmental Variations

Our configuration selection algorithm ranks all the possible configurations based on their score in the previous communications and uses this ranking to suggest a configuration for next backscatter communication. So, the underlying assumption for this algorithm is that the configuration scores do not change rapidly over time. The configuration scores depend on the reflection characteristics of the objects in the environment surrounding the radio units and would change as the objects in the environment move. We perform six different tests to gain a better understanding of the effect of different environmental changes on the quality of the backscatter link. In each test, we let the system run for one minute. Then, we introduce a series of changes in the environment over the next minute and record the backscatter success rates over every second. The results are shown in Fig. 12.

In the first experiment (Fig. 12(a)), we ask two individuals to walk randomly in the space between TRXs and the tag. The average success rate before the event at the tag location is 75.2%, and it falls to 50% after the individuals have started to walk. We do not see any improvement over the next minute. The result is still better than a random



(a) Two individuals walk randomly between TRX1, tag, and TRX2..

(b) Two indivisuals stand 50cm away from tag and TRX1.

(c) Tag is moved 10cm at t=2s and returned to its place at t=32s.

(d) A person waves 10cm away from the receiver antenna.

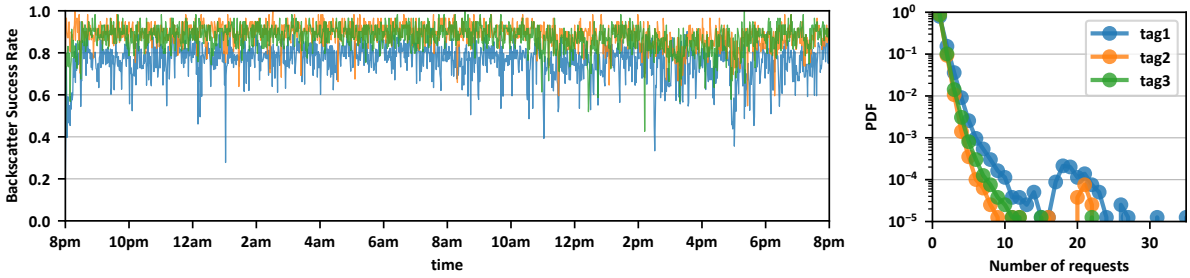
(e) Two full HD videos are streamed to devices 1m away from the receiver.

(f) Tag is turned on and off every 5 seconds.

Fig. 12. Effect of six different environmental variations on the backscatter success rate.

configuration selection (24.3%). A possible explanation for the outcome is that the random movement blocks some signal paths at each instance of time while some other ones remain unaffected. In the second experiment (Fig. 12(b)), two persons block the line of sight between the tag and TRX1 by standing 50 cm away in front of the tag and TRX1 antennas. We observe a fall in the success rate to 40%, which is caused by the higher propagation loss due to blocking the line of sight signal path. In the third experiment (Fig. 12(c)), we move the tag 10cm at t= 2s and return it to its place at t= 32s. We see sharp drops in the success rate following each movement, but the system adapts to the new environment following each event. In the fourth experiment (Fig. 12(d)), a person waves 10 cm away from the receiver antenna. The effect is similar to the first experiment, but since the moving object is smaller (a person's hand), we see a smaller drop in the backscatter success rate. In the fifth experiment (Fig. 12(e)), we place two laptops 1 meter away from the receiver antenna and set them to stream full HD videos using their WiFi cards. We use laptops that only support the 2.4 GHz ISM band. The access point is 2 meters above the receiver. We can see an initial drop in the success rate due to added interference affecting some of the good configurations, which have frequency overlap with the access point frequency band. The system recovers from this and maintains an average 70% success rate over the next 50 seconds of the test. In the final test (Fig. 12(f)), we turn the tag on and off every 5 seconds. The system quickly achieves a high success rate following each turn on, which demonstrates that the system could maintain the configuration ranking for short blockage events. These experiments were only a small fraction of all the possible environmental changes that can happen in real environments. The system continues to receive backscatter packets in all cases, which shows its resilience to environmental variations.

As we discussed earlier, the *Relacks* backscatter system relies on channels with lower propagation loss for backscatter communication. In section 4.2, we show that increasing the time duration between backscatter packets does not change the system performance in a static environment. In such cases, a fixed set of configurations can be used, and the TRXs do not need to share channel metrics very often. However, Fig. 12 shows the channel metrics could change very quickly by a small environmental variation, such as moving the tag by a few centimeters or people walking in the vicinity. In such scenarios, the TRXs have to look for low-loss channels and update configuration constantly. Thus, different backscatter deployments might have very different dynamics, and the optimum rate to share channel metrics has to be selected based on the application and deployment requirements. Increasing the configuration selection update rate increases the overhead. At the same time, it increases the



(a) The average backscatter success rate at each minute.

(b) Distribution of communication attempts before receiving one successful backscatter packet.

Fig. 13. Results of a day-long backscatter communication test with three tags in a 6.4m x 9.4m area.

likelihood of successful communication. A unified approach that provides the optimum update rate for a variety of different environment variations is a subject for future research.

4.5 Long Term Case Study

We set up a day-long test to evaluate the effect of everyday activities on the reliability of the system. We place three tags in the room shown in Fig. 5(a) and program the controller to send a communication request to TRX1 as described in section 3 till it receives one successful backscatter packet from each tag. Once the controller receives one packet from each tag, it waits for one second and repeats the operation after that. We start the test at 8 pm and end it at 8 pm the next day. The room is shared between several people, with 4 people present in the room on average between 8 am to 7 pm. People walk in and out, listen to music on WiFi, and use a microwave oven in the room several times during the test. We use coded PHY packets with the proposed configuration selection algorithm.

The average backscatter success rate at every minute of the test for each tag is plotted in Fig. 13(a). Tag 1, 2, and 3 have average success rates equal to 77.1%, 88.8%, and 87.2%, respectively. We see multiple sudden drops in the minute by minute success rate for each tag. The drops are more frequent during the day than the night, which suggests they could be related to human activities. The system recovers and continues to operate at high success rates shortly after each drop. Fig. 13(b) shows the number of communications requests the controller sends out to get one successful packet. The average number of requests to get one successfully received packet is 1.297, 1.124, and 1.146 for tags 1, 2, and 3 respectively, with 99% of packets received with in the first three attempts. We can identify a second peak at around 20 requests for tag 1 and 2 that has happened multiple times during the test. The source of this second peak is not clear to us.

4.6 Multiple Test Locations

We test the *Relacks* backscatter system in several different indoor environments, including a $48m^2$ build space, a $50m^2$ classroom and a $45m^2$ conference room, to ensure its proper operation in different conditions. In each room, we place the TRXs on two opposite sides of the room, and the test procedures are as described in section 4.1. The average backscatter success rate for both PHYs and three different configuration selection algorithms is plotted in Fig. 14.

The proposed configuration selection algorithm using both antennas on each TRX and coded PHY packets achieves 83% backscatter success rate across all three environments with an average 6% standard deviation. The low standard deviation highlights uniform coverage over the entire area. By limiting each TRX to one antenna, we see a slight drop in the average backscatter success rate and an increase in the standard deviation in two

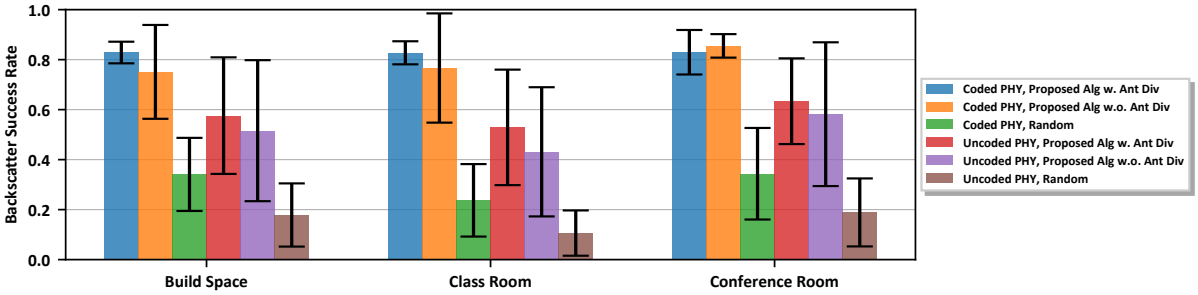


Fig. 14. Testing the Relacks backscatter system in 3 different locations.

of the three environments. Similar to the results discussed in section 4.1, the uncoded PHY measurements on average have lower success rates with a higher variation. In the three tested environments, we see a 2.7 × boost in the backscatter success rate by following the proposed configuration policy compared to random selection for coded PHY and a 3.67 × improvement for uncoded PHY.

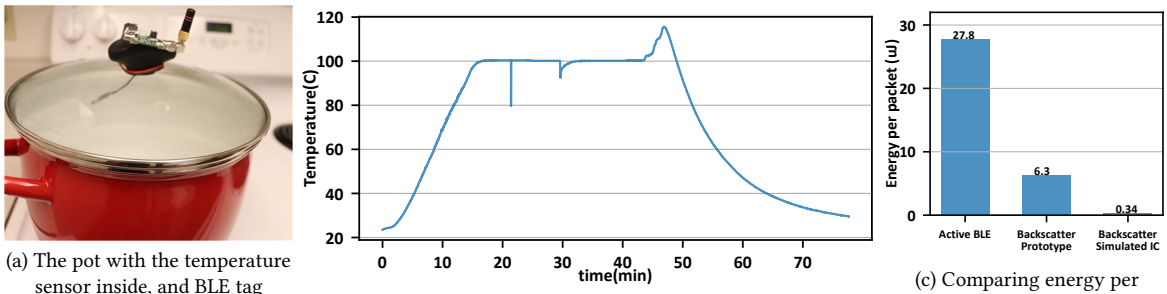
4.7 Application Deployment

Backscatter tags are cheaper than active radios [47] and can bring low-cost wireless connectivity to billions of devices around us to make many new applications possible. We show one sample application using the *Relacks* backscatter communication system. We connect an LMT86 temperature sensor to our tag and read the temperature inside a boiling pot. The data is useful to introduce new cooking recipes or add safety features to the pots. The BLE TRXs are placed on a refrigerator and a home assistant device 2 and 4 meters away from the pot. The pot is filled with water and placed on top of an electric stove. The tag is connected to the pot lid outside the pot. The temperature sensor is inside the pot, and it is connected to the tag with three wires, as shown in Fig. 15(a). We program the controller to send communication requests until it receives a successful backscatter packet. Then it waits for 1 second and repeats the same procedure. We continue the test until all the water inside the pot evaporates, and the temperature rises to 105 degrees Celsius. At this point, we manually turn off the electric stove. The temperature continues to rise to 118 degrees and falls after that. The measured temperature inside the pot is plotted in Fig. 15(b). We measure an average 73.2% packet success rate while two persons freely move around the kitchen during the test.

We compare the average energy required to transmit one successful packet with CC2640R2F BLE transceiver [2], our prototype BLE backscatter tag, and the simulated IC design for BLE backscatter tag in Fig. 15(c). We use equation 5 to calculate the energy per successful packet.

$$E_{tot} = P_{tx} * T_{tx} * \frac{1}{successrate} \tag{5}$$

We set the CC2640R2F to read the LMT86 temperature sensor and transmit one BLE 1M PHY packet. The entire process to exit sleep mode, activate the sensor, read the output with ADC, and transmit a BLE packet takes 2.9 ms. The average measured current consumption is 3.2 mA from a 3V supply. We assume a success rate equal to 1 for the active BLE packet, which brings the total energy consumption to 27.8 uj. Our prototype backscatter tag consumes an average 1.1 mW over 4.2 ms to send one BLE coded PHY packet. The average power consumption is lower than what measured in section 2 because the backscatter communication only takes 1 ms, and the remaining time is spent on waiting for the oscillator to settle. We use the same success rate as calculated in our measurement. The total energy consumption to transmit one backscatter packet with our prototype is 6.3 uj. Based on the simulations, we project the BLE backscatter integrated circuit consumes 60 uW over the same period with equal success rate to our prototype. Thus, we expect 0.344 uj total energy consumption, which is 80 times less than the active radio.



(a) The pot with the temperature sensor inside, and BLE tag outside it.

(b) The recorded temperature.

(c) Comparing energy per successful packet.

Fig. 15. Measuring the temperature inside a boiling pot with the BLE backscatter tag.

5 RELATED WORKS

This work is related to prior efforts on backscatter communication systems. Long-range backscatter systems with communication range beyond hundred meters are introduced in [38, 47, 49] to cover an entire house or office using a single transmitter and receiver. These works achieve an extended communication range due to the lower sensitivity of the receiver and favorable propagation characteristics in the 900 MHz frequency band. In this work, we limit the scope of the backscatter system to a single room, which allows us to operate with a lower excitation power and a higher data rate at 2.4 GHz ISM band. Backscatter systems based on 2.4 GHz ISM band standards such as Bluetooth [17, 25, 52], WiFi [9, 25, 29, 52, 53, 55] and Zigbee [32] have introduced and achieved room level communication ranges. In this work, we extend the application of such systems by adding a method for reliable backscatter communication in a multipath-rich indoor environment.

The fading effect in indoor environments is a well-known issue [13, 19, 48]. Since we are dealing with two consecutive propagations in backscatter communication, the multipath fading could severely limit the reliable communication range. [22, 30, 31] provide detailed measurement results on the effect of multipath fading on backscatter links in 900 MHz, 2.4 GHz, and 5.8 GHz ISM bands. [26] exploits a multi-carrier backscatter system and channel diversity to communicate with multiple tags in parallel and increase the throughput. It uses a software-defined-radio based transmitter and receiver for that purpose. [51] and [21] use link metrics, bitrate selection, and channel selection algorithms to increase the throughput of the backscatter RFID system. These works are limited to RFID applications, where a single base unit provides the excitation signal and receives the reflections. [40] highlights the multipath fading issue for backscatter communication inside an animal cage environment and relies on a one-time calibration and frequency planning between three BLE channels to improve the link reliability. Although [40] shows the importance of frequency diversity, it does not present a dynamic approach to handle environmental variation in a larger space. [49] uses multiple transmitters and receivers at different frequencies to overcome the frequency-selective multipath. This architecture occupies multiple channels and requires an additional set of transmitters and receivers for each added frequency diversity.

6 CONCLUSION AND FUTURE DIRECTIONS

We present an efficient backscatter system that can provide reliable coverage in indoor environments. The backscatter system relies on a closed-loop configuration selection algorithm that uses frequency, antenna, and wake-up source diversity to maximize the success rate of backscatter packets. We prototype a complete backscatter system with BLE transceivers and BLE compatible tags and use them in evaluating our system. We discuss FCC compatibility, BLE compatibility, and synchronization overhead of the proposed method. The measurement results show more than 80% average backscatter success rate over the tested points inside multiple indoor environments, 2.7 times improvements over random selection baseline. We show that the proposed method is resilient against

environmental variation and tag antenna orientation. Finally, we test our system for an extended period and provide a sample application to show its performance in real-life scenarios.

Next, we outline topics for future research.

Excitation power tuning. Backscatter links received signal strength varies a lot depending on the relative distance of RF excitation source, tag, and receiver. Although strong RF excitation signal might be essential in unfavorable conditions, it would generate a substantial interference to other wireless communication systems in its vicinity. The accurate control of excitation power allows minimizing interference between different systems operating concurrently and enabling the vision of connecting billions of devices. A closed-loop system such as the one explained in this work can be used to enable excitation power control based on the channel metrics extracted at the receiver.

Low power wake up radios. Backscatter technique reduces the energy required to transmit data from low-power sensor nodes, however these nodes also need receivers to get network commands and settings. Although the passive envelope detection technique has low power consumption, it only allows one downlink communication from base units to a tag at a time. Low power wake-up radio architectures is an open research problem.

Networking backscatter devices. The frequency and timing synchronization for the backscatter operation requires exchanging control packets between the excitation source, tag, and receiver, which increases the communication overhead. We believe there is a research opportunity in maximizing the overall application throughput by proposing algorithms that minimize the synchronization overhead.

Human Sensing. In this paper, we show the effects of human movement on backscatter links. We believe this information, along with other channel parameters, can help to determine the human presence in target zones.

Energy harvesting. The backscatter tag design presented in this work can be powered using coin cell batteries or harvesters that convert motion, vibration, light, EM radiation, or thermal energy into electrical energy. Low cost, efficient energy harvesting is of great interest to achieve ubiquitous connectivity.

Localization. Reliable backscatter communication can also bring indoor localization to a new category of small unplugged objects. Several TRX units can work cooperatively to localize low power tags. Similarly, several fixed tags and one or more active TRX could provide sufficient information to localize another active TRX. In all these scenarios, one has to consider reliability, accuracy, and latency tradeoffs for different solutions.

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